



**Verified Carbon
Standard**

UZUNDERE I 63.0 MW HYDROELECTRIC POWER PLANT PROJECT, TURKEY

Document Prepared by Sekans Enerji

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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey, which is developed by KARADENIZ HES ELEKTRİK A.Ş. (hereafter referred to as the “project owner”) is a run-of-river hydropower project located in Rize Province, Turkey. The project is on the Uzundere creek. The altitude is 817.59 m at powerhouse location.

The Project has been implemented and operated by KARADENIZ HES ELEKTRİK A.Ş. The project aims to generate electricity from hydropower and feed it to the national electricity grid.

Total installed capacity of the project is 63 MWe, consisting of 2 turbines, with an estimated power supply to the grid of 156,205 MWh per annum. The project is in implementation since May 27th, 2010, when one of the two vertical shaft Pelton turbines in the plant commences electricity generation. The starting date of the project activity for the other turbine has been November 7th, 2010. The estimated amount of GHG emission reduction is 87,318 tonnes CO₂e per year in the registered PDD. During its operation during this monitoring period, the actualized net electricity generation is 317,508.93 MWh. By displacing equivalent amount of electricity consumed from the national grid, which is mostly comprised of fossil fired power plants, the actual emission reduction has been calculated as 177,485 tonnes CO₂ for this monitoring period.

The implementation of the project consists of two diversion weirs, where water from the river is diverted into conveyance pipes; a conveyance tunnel with a length of 11.324 km and powerhouse with two Pelton type vertical turbines. Each turbine has a capacity of 31.534 MWm / 31.076 MWe.

Total installed capacity in the powerhouse is 63.068 MWm / 62.152 MWe). There are two generators attached to the facility. Generators have power factor of 0.9 d/d, a frequency of 50 Hz and an output of 34.5 MVA.

The construction of the project started in 01/04/2010 and the project was operational on 27/05/2010.

The project has produced positive environmental benefits as displacing the electricity generated by fossil fuel fired power plants by utilising the renewable resources so as to avoid environmental pollution and GHG emissions.

Please see the vintage-based comparison of expected and achieved emission reductions for this monitoring period:

Table 1 - Expected vs achieved emission reductions

Vintage	Period	Total Days	Amount achieved during this monitoring period (tCO ₂ e)	Amount estimated ex ante (tCO ₂ e)	Difference (tCO ₂ e)	Difference (%)
2017	01.12.2017 - 31.12.2017	31	2,957	7,416	-4,459	-60.1%
2018	01.01.2018 - 31.12.2018	365	75,863	87,318	-11,455	-13.1%
2019	01.01.2019 - 31.12.2019	365	68,882	87,318	-18,436	-21.1%
2020	01.01.2020 - 26.05.2020	147	29,783	35,166	-5,383	-15.3%
Total		908	177,485	217,218	-39,733	-18.3%

Please see the Monitoring Timeline of the Project Activity as below:

Table 2 - Monitoring timeline of the Project Activity

Date (Period)	Activity
01/04/2010	Issuance of EIA Certificate
01/04/2010	Construction Start Date
27/05/2010	Commissioning of the Project
27/05/2010 - 01/12/2011	1 st Monitoring Period
02/12/2011 - 30/11/2017	2 nd Monitoring Period
01/12/2017 – 26/05/2020	3 rd Monitoring Period

Please see the technical description of the Project Activity through Table 3:

Table 3 - Technical description of the Project Activity¹

¹ The photographic evidence of turbine name plates is available to the VVB.

Turbines	
Turbine Brand	ALSTOM POWER SYSTEMS HYDRO-IBERICA
Turbine Type	Vertical Pelton
Turbine Serial Numbers	2073 / 2074
Number of units	2
Turbine Rotations	750 rpm
Water flow rate	749,41 m ³ /s
Net head at 9.00 m ³ /s	66.22 m
Rated power	31.534 MW

Generators	
Generator brand	ALSTOM POWER SYSTEMS HYDRO-IBERICA
Generator type	SAV 217/120/8, 3 phased
Number of units	2
Nominal power of a unit	34500 kVA
Voltage	11000 V
Frequency	50 Hz
Power factor	0.96
Revs per minute	750

1.2 Sectoral Scope and Project Type

The Project is a voluntary emission reduction project. It is not a part of any GHG program approved by the VCS Board (e.g., UNFCCC CDM, UNFCCC JI, or the Californian Climate Action Registry). Its carbon reductions are not registered under any program.

The Project is not a grouped project or a debundled part of a larger project. The project activity does not use any Official Development Aid (ODA) finance.

According to UNFCCC sectoral scopes definition for CDM projects, the Project Activity is included in the Sectoral Scope 1, category “Energy industries (renewable - / nonrenewable sources)”. The project is a single green field investment and is not part of a project group or bundle.

1.3 Project Proponent

Organization name	Karadeniz HES Elektrik A.Ş.
Contact person	Orçun Altınbaş
Title	ARGE Engineer
Address	Fahrettin Kerim Gökay cad. No:36 Altunizade 34662 Üsküdar – İstanbul - Turkey
Telephone	0090-216-5442400
Email	orcun.altinbas@iltekenerji.com.tr

1.4 Other Entities Involved in the Project

Organization name	Sekans Enerji Ltd. Şti
Role in the Project	Consultant
Contact person	Sıla Duran
Title	Consultant
Address	Emniyet Evleri mah. Eski Büyükdere cad. No: 1/1 İç Kapı No: 1B04 Kağıthane/İstanbul
Telephone	+90 532 438 30 29
Email	sil@sekansdanismanlik.com

1.5 Project Start Date

The project start date is 27/05/2010² as it is the date of commissioning date of the project.

² The Ministry Acceptance Protocols are available to the VVB.

1.6 Project Crediting Period

Two times renewable crediting period of 10 years 0 month shall apply. The Project's total crediting period is: 27/05/2010 – 26/05/2020 (both days are included).

1.7 Project Location

The project is located in the Black Sea region of Turkey, within the Province of Rize. The project activity is located on the Uzundere creek. The altitude at the powerhouse location is 817.59 m. The project site was treasury land. The project's geographical location is 40°50'16"N - 40°54'04"N and 40°45'4"E - 40°49'42"E.



Figure 1. Location of the Project in Turkey Map

1.8 Title and Reference of Methodology

The emission reductions of the Project were calculated in accordance with the approved consolidated baseline and monitoring methodology ACM0002, Version 11: "Consolidated baseline methodology for grid-connected electricity generation from renewable sources".

Also, to prove additionality and to calculate the grid emission factor,

“Tool for the demonstration and assessment of additionality, Version 05.2 and

“Tool to calculate the emission factor for an electricity system”, Version 02 are used.

1.9 Participation under other GHG Programs

The project has never been included in an emissions trading program or any other binding limits. It has neither received any kind of environmental credits nor been registered under any other GHG programs.

1.10 Other Forms of Credit

The project has not created another form of credit.

1.11 Sustainable Development Contributions

The project produces electricity from renewable energy sources using hydro as the power source and to contribute to Turkey's growing electricity demand through a sustainable and low carbon technology. The project displaces the same amount of electricity generated by the grid dominated with fossil fired power plants. The project is expected to generate 156,205 MWh annually. The project contributes to the Sustainable Development Goal, Affordable and Clean Energy. During this monitoring period, the project has generated 317,508.93 MWh as contributing to SGD 7 Affordable and Clean Energy.

The annual emission reduction estimated by the project is 87,318 tonnes CO₂eq, approximately. While this amount of emissions is mitigated, technology transfer is also realized as benefitting from hydropower. During this monitoring period, the project has contributed to SDG 13 Climate Action as reducing 177,485 tonnes CO₂.

The project contributes to improve the environmental situation in the region and in the country as avoiding fossil fuel-based electricity enhanced the air quality and help to reduce the negative effects on the climate. Through renewable technologies and hydro-based electricity sustainable and climate friendly development is promoted. The project contributes to the Sustainable Development Goal, Climate Action.

During construction and operational period, the project has created employment opportunities for the local community. As contributing to SGD 8 Decent Work and Economic Growth, 16 employees are working for the project activity.

The project contributes the economic development of the region by providing sustainable energy resources. The project provides workers with a safe and healthy work environment and is not complicit in exposing workers to unsafe or unhealthy work environments. Thus,

the project contributes to the Sustainable Development Goal, Decent Work and Economic Growth.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2.	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	The project produces electricity from renewable energy sources using wind as the power source and contributes to Turkey's growing electricity demand through a sustainable and low carbon technology. During this monitoring period, the project has prevented 317,508.93 MWh from fossil fuels.	Preventing the generation of 5,467,175 MWh electricity generation from fossil fuels.

2)	8.2	Job creation for local people in a high value added sector	Implemented activities to decrease/increase	During construction and operational period, the project has created employment opportunities for the local community. The project contributes to the economic development of the region by providing sustainable energy resources. As contributing to SGD 8 Decent Work and Economic Growth, 16 employees are working for the project activity.	As contributing to SGD 8 Decent Work and Economic Growth, 16 employees are working for the project activity
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By producing electricity from hydro power, the project activity has prevented the release of 177,485tonnes of carbon into the atmosphere during the reporting period.	Preventing the release of 3,056,130 tonnes of carbon into the atmosphere.

2 SAFEGUARDS

2.1 No Net Harm

There is not any negative environmental or socio-economic impact.

The project paid extra attention to maintaining social and environmental stability:

- The construction followed strict noise and dust standards to prevent any inconvenience to communities. Same standards are followed during the operation of the plant.
- The construction followed various strict standards for labor health and site safety.
- No pollution from any source is released into the river from the operation of the plant. Hazardous wastes are handled appropriately in closed containers and transported by licensed transporters to the licensed processing and disposal facilities.
- Wastewater is collected through within the septic tank and is transferred through the sewage truck³.
- Since the Project Activity is outside the municipal boundaries, the domestic wastes are taken by the vehicles of provincial special administration and taken to recycling site of Rize Municipality⁴.
- Regarding the monitoring of “Water quality and quantity”, “Biodiversity and aquatic life”, streamgauge gauging station is built for the measurement of flow to assess the environmental flow. General directorate for state hydraulic works executes the monitoring and the Project complies with the regulations.⁵

2.2 Local Stakeholder Consultation

According to the regulation declared by the Ministry of Environment and Forestry and published on the Official Gazette (dated 17th July 2008, numbered 26939), projects which an EIA is not required from are not subject to an obligation to hold Stakeholder Meetings. Therefore, a stakeholder meeting has not been planned. Nevertheless, Mukhtars of local villages have been interviewed and continuously informed about the Project for understanding the general perspective about the project.

³ The wastewater transfer receipts are available to the VVB

⁴ The records are available to the VVB.

⁵ Flow measurement (lifeline water) reports are available to the VVB.

In summary, local communities are supporting the project and they believe that it creates a momentum for local sustainable development. The Project has resulted in the creation of new jobs in the village, improving life standards in the vicinity.

There was no specific environmental problem that concerned the locals more than others, and environmental issues need to be solved by the government. All people considered clear signs of climate change in the region in recent years.

The contact information of the plant responsible exist at the Mukhtar, the project owner and local community are always in touch. The project owner regularly checks with the Mukhtar if any complaint or a request exists. Any complaint or need from the local community could directly be received by the project owner and appropriate contributions or improvements are made to the local community.

An announcement stating the contact details of the Project relevant staff in case of any complaint by close surroundings has been put up on the village mosque to be seen by the villagers. Regarding the monitoring period, a signed letter by the Mukhtar has been provided.

2.3 AFOLU-Specific Safeguards

N/A

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The description of the implementation and operational status of the project and the installed technology(ies), technical process and equipment are detailed in Section 1.1.

Other entity involved in the Project was GAIA Finansal Danismanlik, Turkey in the project description. However, it's been changed as Sekans Danışmanlık after the first monitoring period.

3.2 Deviations

3.2.1 Methodology Deviations

There are no methodology deviations applied during this monitoring period.

3.2.2 Project Description Deviations

The description of the implementation and operational status of the project and the installed technology(ies), technical process and equipment are detailed in Section 1.1.

The project Proponent name has been changed as Karadeniz HES Elektrik A.Ş. after the first monitoring period⁶. Other entity involved in the Project was GAIA Finansal Danismanlik, Turkey in the project description. However, it's been changed as Sekans Danismanlik after the first monitoring period.

Other entity involved in the Project was Global Tan Enerji Ltd. in the project description. However, it's been changed as Sekans Enerji Ltd. Şti. after the 1st periodic monitoring period.

3.3 Grouped Projects

There is not a grouped project.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF_{grid, CM, y}
Data unit	tCO ₂ e/MWh
Description	Combined margin CO2 emission factor for grid connected power generation in year
Source of data	Registered PDD
Value applied	0.559
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Data used for calculation of baseline emissions
Comments	-

⁶ The screenshot of the project license from the website of EMRA is available to the VVB.

Data / Parameter	FC_{i,y}
Data unit	ton or 1000 m ³
Description	Amount of fossil fuel type i consumed in the project electricity system
Source of data	As per the "Tool to calculate the emission factor for an electricity system"
Value applied	Please refer to Annex 3 in the PDD
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Data used for calculation of baseline emissions
Comments	-

Data / Parameter	NCV_{i,y}
Data unit	TJ/kt or TJ/m ³
Description	Net calorific value of fossil fuel type i in year y
Source of data	As per the "Tool to calculate the emission factor for an electricity system"
Value applied	Please refer to Annex 3 in the PDD
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Data used to calculate the combined margin emission factor
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	EG_{facility,y}
------------------	--------------------------------

Data unit	MWh																		
Description	Annual net electricity generated by the Project Activity																		
Source of data	Measured by TEIAS-approved measurement devices and attained from the EPIAS ⁷ website, which is also used by TEIAS for invoicing.																		
Description of measurement methods and procedures to be applied	The Project Activity is connected to the Çayeli Transformer Station via a 154 kV transmission line. There are two metering instruments (power meters). These devices measure the net electricity supply to the national grid by the Project Activity, all losses before this point are on account of the project participant. All metering instruments, which continuously monitor and measure the net electricity delivered by the Project Activity, are sealed and only accessible by TEIAS personnel. Official TEIAS data from the EPIAS web site was used for calculating EG _{Facility, y} .																		
Frequency of monitoring/recording	Monthly readings																		
Value monitored	5,289,88 MWh for 2017 135,713.03 MWh for 2018 123,225.25 MWh for 2019 53,280.77 MWh for 2020																		
Monitoring equipment	The net electricity is measured continuously by four electricity meters at the grid interface and recorded monthly. Power meters: <table><tr><th></th><th>Main Meter</th><th>Spare Meter</th></tr><tr><td>Name</td><td>Ana Sayaç</td><td>Yedek Sayaç</td></tr><tr><td>Brand</td><td>EMH</td><td>EMH</td></tr><tr><td>Serial Number</td><td>9674605</td><td>9674606</td></tr><tr><td>Latest Test Date of the Meters</td><td>15/10/2020</td><td>15/10/2020</td></tr><tr><td>Accuracy Class</td><td>C-1S</td><td>C-1S</td></tr></table> The calibration date of the meters is 15.10.2020. Power meters that were changed on 15.10.2020:		Main Meter	Spare Meter	Name	Ana Sayaç	Yedek Sayaç	Brand	EMH	EMH	Serial Number	9674605	9674606	Latest Test Date of the Meters	15/10/2020	15/10/2020	Accuracy Class	C-1S	C-1S
	Main Meter	Spare Meter																	
Name	Ana Sayaç	Yedek Sayaç																	
Brand	EMH	EMH																	
Serial Number	9674605	9674606																	
Latest Test Date of the Meters	15/10/2020	15/10/2020																	
Accuracy Class	C-1S	C-1S																	

⁷ PMUM has been replaced by EPIAS as of 01/09/2015 in Turkey during the verification period. Retrospective data is accessible via EPIAS.

		Main Meter	Spare Meter
	Name	Ana Sayaç	Yedek Sayaç
	Brand	ACTARIS	ACTARIS
	Serial Number	53064430	53064429
	Accuracy Class	0.2S	0.2S
Date of the First Index Protocol is 08/02/2010. ⁸			
Latest meter test dates are 20/11/2017 and 06/11/2018.			
QA/QC procedures to be applied	- Spare meters are used for crosschecking the accuracy and all meters are calibrated if required. - The EPIAS records are considered as the main source for the net electricity and the values are crosschecked with the Meter Reading Forms. - TEIAS is responsible for calibration and maintenance of the devices. The periodic calibration or maintenance is under the responsibility of TEIAS and has been fixed as once in 10 years⁹. Since TEİAŞ meters are sealed by TEİAŞ, the project proponent cannot intervene with the devices. The periodic tests are executed on annual basis.¹⁰		
Purpose of the data	Calculation of net electricity supplied to the grid and thus baseline emissions		
Calculation method	Direct continuous measurement		
Comments	-		

Data / Parameter	CAP _{PJ}
Data unit	MW
Description	Installed capacity of the hydro power plant after the implementation of the project activity
Source of data	Project site

⁸ Protocol has been provided to the VVB.

⁹ <http://www.mevzuat.gov.tr/Metin.Aspx?MevzuatKod=7.5.6381&MevzuatIliski=0&sourceXmlSearch>
¹⁰ Test protocols are available to the VVB.

Description of measurement methods and procedures to be applied	Supplier information on the equipment
Frequency of monitoring/recording	Yearly
Value monitored	63.00
Monitoring equipment	-
QA/QC procedures to be applied	Supplier information on the related equipment and the existence of the equipment will be checked. Also, this information may be checked from the EMRA website.
Purpose of the data	Data used to calculate the power density of the project activity
Calculation method	-
Comments	-

Data / Parameter	A_{PJ}
Data unit	m ²
Description	Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full
Source of data	Project site
Description of measurement methods and procedures to be applied	Measured from topographical surveys, maps, satellite pictures, etc.
Frequency of monitoring/recording	Yearly
Value monitored	7,275,54 ¹¹
Monitoring equipment	Topographical surveys, maps, satellite pictures

¹¹ Please see the registered PD for the calculation. Calculation is also available in the ER Calculations file.

QA/QC procedures to be applied	The readings will be done during the period when the water flow is made to have the largest reservoir area
Purpose of the data	Data used to calculate the power density of the project activity
Calculation method	-
Comments	-

4.3 Monitoring Plan

The objective of the monitoring plan is to ensure the complete, consistent, clear, and accurate monitoring and calculation of the emission reductions during the whole crediting period. The Project proponent is responsible for the implementation of the monitoring plan.

The Project Proponent is responsible for the overall management of the monitoring procedures including recording, data collection, calculating emission reductions and project emissions.

Please see below the management structure for the plant operation:

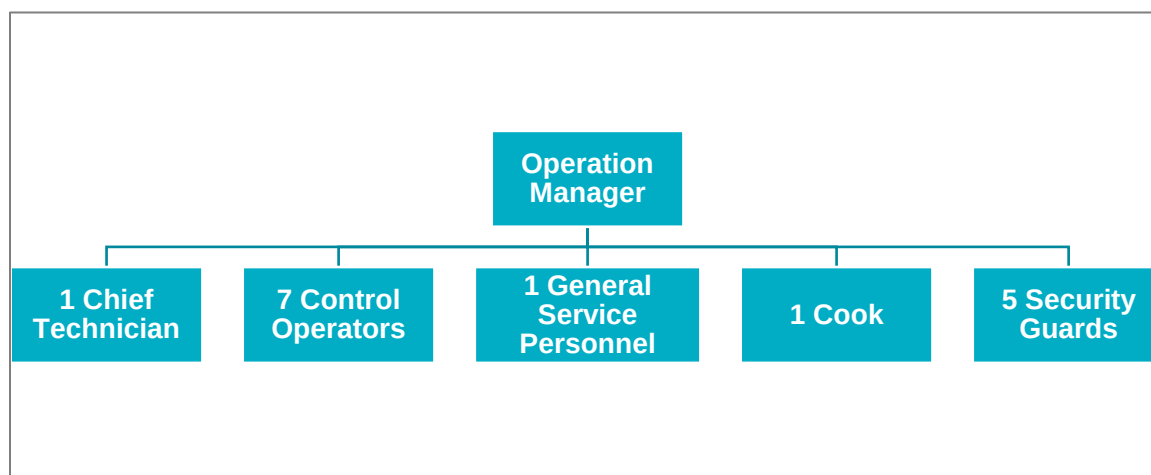


Figure 2. The management structure for the plant operation

The Operation Manager is responsible for the whole management of the Project Activity and the chief technician and control operators are responsible for the implementation of the Project Activity. In addition to the operators, there is one general service personnel, one cook and five security guards. In total, sixteen employees are working for the Project Activity.

Two power meters are installed at the grid interface of the project. One of them is the main meter, and the other one is the spare meter of the main meter for cross-checking. Both meters are jointly inspected and sealed in order to be protected from interference by any of the parties. Both the main and secondary meters are owned and installed by the grid operator (TEIAS). The grid operator's is the only one authorized to deal with fixing, calibrating, or changing the meters, which is being done either by the grid operator or by a company authorized by the grid operator. During this monitoring period, there were no records of meter failure. In case of any urgent case TEIAS contacts the Plant Manager. Since the meters are within TEIAS' province, TEIAS executes all the procedures for handling non-conformities. Therefore, the Project Participant does not have any internal auditing for this purpose.

TEIAS is performing remote reading of the meters and monthly power meter readings are the basis for monitoring net electricity fed into the grid. A measuring protocol is prepared including day, peak and night hour electricity generation by the project owner and approved by governmental officers at the end of each month.

The primary source is the Meter Reading Forms. Net electricity exported is crosschecked with EPIAS records. EPIAS is the financial settlement center of TEIAS¹². The Meter Reading Forms are filled by the project owner and approved by the governmental officers. Additionally, the remote reading by the governmental body is also available. The website of EPIAS is accessible to Project Proponent with their unique user ID and password. Once accessed, the Project Proponent is able to call electricity generation and consumption reports of their own projects. The same reports are used by the Project Proponent for invoicing TEIAS. The electricity generation data is reported monthly basis.

The third data to be monitored is the reservoir area of the Project Activity. The reservoir area corresponding to maximum operational level has been determined as a certain value according to the topographical maps.

Quality Control and Assurance (QC/QA)

Both the main and spare (control) meters are owned and installed by the grid company. All meters are jointly inspected and sealed in order to be protected from interference by any of the parties.

The grid operator's Metering Officer should be notified of any failure of one of the meters. He is the only one authorized to deal with fixing, calibrating, or changing the meters, which is being done either by the grid operator or by a company authorized by the grid operator. During this monitoring period, there were no records of meter failure. The data is hourly recorded by the personnel on the plant. TEIAS Load Dispatching Center also executes

¹² PMUM has been replaced by EPIAS as of 01/09/2015 in Turkey during the verification period. Retrospective data is accessible via EPIAS.

remote reading during 24-hour period. At the end of the month, the readings are taken to the center to be compared and approved. The data could be remotely accessed by TEIAS.

All data collected as part of monitoring is being archived electronically by the Project Proponent and be kept at least for 2 years after the end of the last crediting period.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The baseline emission BE_y (tCO₂e) during the monitoring period results from:

$$BE_y = EG_{PJ,y} * EF_{grid, CM, y}$$

where:

BE_y	Baseline emissions in year y (tCO ₂ e/y)
$EG_{PJ,y}$	Quantity of net electricity supplied to the grid as a result of the implementation of the CDM project activity in year y (MWh)
$EF_{grid, CM, y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated by using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO ₂ /MWh)

$EF_{grid, CM, y}$ value is fixed as 0.559 tCO₂/MWh for during the crediting period. Table 4 shows the baseline emissions for the verification period:

Table 4 - Baseline emissions

Month	(A) Electricity supplied to the grid [MWh]	(B) Electricity consumption from the grid [MWh]	(C) = (A) - (B) EG (ID 8) Net electricity supplied to the grid [MWh]	EF [tCO ₂ /MWh]	Baseline emission: ER = EG * EF [t CO ₂ -eq]
Dec-17	5,321.61	31.73	5,289.88	0.559	2,957

Jan-18	4,409.00	36.74	4,372.26	0.559	2,444
Feb-18	3,934.96	31.79	3,903.17	0.559	2,182
Mar-18	8,652.20	25.84	8,626.36	0.559	4,822
Apr-18	14,215.09	5.80	14,209.29	0.559	7,943
May-18	35,505.72	0.52	35,505.20	0.559	19,847
Jun-18	24,917.14	0.75	24,916.39	0.559	13,928
Jul-18	11,211.20	5.61	11,205.59	0.559	6,264
Aug-18	8,920.43	18.43	8,902.00	0.559	4,976
Sep-18	5,607.83	21.03	5,586.80	0.559	3,123
Oct-18	7,053.63	22.06	7,031.57	0.559	3,931
Nov-18	5,999.98	27.38	5,972.60	0.559	3,339
Dec-18	5,514.88	33.08	5,481.80	0.559	3,064
Jan-19	4,019.22	39.63	3,979.59	0.559	2,225
Feb-19	3,201.42	34.58	3,166.84	0.559	1,770
Mar-19	3,865.21	40.19	3,825.02	0.559	2,138
Apr-19	8,174.09	21.75	8,152.34	0.559	4,557
May-19	30,138.86	1.51	30,137.35	0.559	16,847
Jun-19	25,047.76	1.33	25,046.43	0.559	14,001
Jul-19	13,380.01	6.03	13,373.98	0.559	7,476
Aug-19	8,944.85	11.48	8,933.37	0.559	4,994
Sep-19	9,026.70	11.92	9,014.78	0.559	5,039
Oct-19	7,716.81	13.25	7,703.56	0.559	4,306
Nov-19	5,587.40	18.98	5,568.42	0.559	3,113
Dec-19	4,355.31	31.74	4,323.57	0.559	2,417
Jan-20	3,583.80	37.39	3,546.41	0.559	1,982
Feb-20	3,138.38	40.06	3,098.32	0.559	1,732
Mar-20	7,071.15	26.35	7,044.80	0.559	3,938
Apr-20	10,222.00	17.91	10,204.09	0.559	5,704
1/05/2020- 26/05/2020	29,389.17	2.02	29,387.15	0.559	16,427
2017 Vintage (01.12.2017- 31.12.2017)	5,321.61	31.73	5,289.88	0.559	2,957
2018 Vintage (01.01.2018- 31.12.2018)	135,942.06	229.03	135,713.03	0.559	75,863
2019 Vintage (01.01.2019- 31.12.2019)	123,457.64	232.39	123,225.25	0.559	68,882
2020 Vintage (01.01.2020- 26.05.2020)	53,404.50	123.73	53,280.77	0.559	29,783

Total	318,125.81	616.88	317,508.93	0.559	177,485
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5.2 Project Emissions

There is no project emission resulting from the reservoir area of the Project Activity as the power density of the project is greater than 10W/m².

Accordingly, The Project Activity's power density was calculated as below:

$$= 63,000,000 \text{ W} / 7,275.54 \text{ m}^{213}$$

$$= 8,659.2 \text{ W/m}^2$$

Therefore, PE=0.

5.3 Leakage

No leakage is to be accounted by the Project Activity. This is in line with the registered PD and applicable methodology ACM0002, Version 11. Therefore, the leakage from the Project Activity is zero.

¹³ Please see the registered PD for the calculation. Calculation is also available in the ER Calculations file.

5.4 Net GHG Emission Reductions and Removals

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$$

Where:

BE_y	Baseline emissions in year y (tCO ₂ e/y)
$EG_{PJ,y}$	Quantity of net electricity supplied to the grid as a result of the implementation of the CDM project activity in year y (MWh)
$EF_{grid,CM,y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated by using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO ₂ /MWh)

Table 5 – Net GHG Emission Reductions and Removals:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2017	2,957	0	0	2,957
2018	75,863	0	0	75,863
2019	68,882	0	0	68,882
2020	29,783	0	0	29,783
Total	177,485	0	0	177,485